Cramér-Rao Bounds on Sparse–Diffuse Channel Estimation: Extended Appendix

Lei Lyu, Maxime Ferreira Da Costa, and Urbashi Mitra

December 23, 2025

Proposition 1. *Assume* $\boldsymbol{n} \sim \mathcal{CN}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_N)$ *and* $\boldsymbol{S}^{\mathsf{H}}\boldsymbol{S} = \boldsymbol{I}_N$. *Let* $\delta = \min(\delta_{\mathrm{s}}, \delta_{\mathrm{s-d}})$. *Furthermore, assume* $\alpha_i \neq 0$ *for all* $i \in [\![1, m]\!]$. *If* $2m + L \leq N$ *and* $N\delta > 2$, *then*

$$K_{\min}|\alpha_i|^{-2}\sigma^2 \leq \frac{\pi^2(N-1)(2N-1)}{3}\mathrm{CRB}(f_i) \leq K_{\max}|\alpha_i|^{-2}\sigma^2, \tag{1a}$$

$$K_{\min}\frac{\sigma^2}{N} \leq \mathrm{CRB}(\alpha_i) \leq K_{\max}\frac{\sigma^2}{N}, \tag{1b}$$

$$K_{\min}\frac{\sigma^2}{N} \leq \mathrm{CRB}(\gamma_r) \leq K_{\max}\frac{\sigma^2}{N}, \tag{1c}$$

for all $i \in [\![1, m]\!]$ *and all* $r \in [\![1, L]\!]$, *where* $K_{\min}$ *and* $K_{\max}$ *are the constants depending on the separation parameter* $N\delta$

$$K_{\min} = \frac{1}{\left(1 + 2(N\delta)^{-1}\right)},$$
$$K_{\max} = \frac{1}{\left(1 - 2(N\delta)^{-1}\right)}.$$

Proof. Under the hypothesis of Proposition 1 the FIM reads $\boldsymbol{J}^{\boldsymbol{\theta}} = \sigma^{-2}\boldsymbol{U}^{\mathsf{H}}\boldsymbol{U}$. To proceed with the proof, we let the scaling matrix $\boldsymbol{T}$ be defined by

$$\boldsymbol{T} = \begin{bmatrix} \sqrt{2}C_N \operatorname{diag}(\boldsymbol{\alpha}) & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0} & N\boldsymbol{I}_{m+L} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & N\boldsymbol{I}_{m+L} \end{bmatrix} \in \mathbb{C}^{(3m+2L)\times(3m+2L)}, \tag{2}$$

where $C_N = \|\boldsymbol{\Lambda}\boldsymbol{a}(f)\|_2 = \sqrt{\frac{\pi^2 N(N-1)(N+1)}{3}}$ for all $f \in [0, 1)$ is the norm of the derivative of the complex exponential atoms. Furthermore, introduce $\boldsymbol{Q} = \boldsymbol{U}\boldsymbol{T}^{-1}$. With the notation in place, the CRB matrix $\boldsymbol{\Theta}_{\mathrm{CRB}}$ writes $\boldsymbol{\Theta}_{\mathrm{CRB}} = \boldsymbol{J}^{\boldsymbol{\theta}^{-1}} = \sigma^2\boldsymbol{T}^{-\mathsf{H}}\left(\boldsymbol{Q}^{\mathsf{H}}\boldsymbol{Q}\right)^{-1}\boldsymbol{T}^{-1}$. The CRB on a specific frequency $\mathrm{CRB}(f_i)$ or sparse channel amplitudes $\mathrm{CRB}(\alpha_i)$ and diffuse channel amplitudes $\mathrm{CRB}(\gamma_r)$ are given by the respective diagonal entries of $\boldsymbol{\Theta}_{\mathrm{CRB}}$.

$$\boldsymbol{\Theta}_{\mathrm{CRB}}|_{j,j} = \sigma^2 T_{j,j}^{-2}\left[\left(\boldsymbol{Q}^{\mathsf{H}}\boldsymbol{Q}\right)^{-1}\right]_{|j,j}. \tag{3}$$

Denote by $\lambda_{\max}(\boldsymbol{M})$ and $\lambda_{\min}(\boldsymbol{M})$ the largest and smallest eigenvalue of a Hermitian positive matrix $\boldsymbol{M}$, respectively. Since $\lambda_{\min}(\boldsymbol{M}) \leq M_{i,i} \leq \lambda_{\min}(\boldsymbol{M})$ for any diagonal entries $M_{j,j}$ of the matrix $\boldsymbol{M}$, (3) induces the frame

$$\sigma^2 T_{j,j}^{-2} \lambda_{\max}(\left(\boldsymbol{Q}^{\mathsf{H}}\boldsymbol{Q}\right)^{-1}) \leq \boldsymbol{\Theta}_{\mathrm{CRB}}|_{j,j} \tag{4}$$

$$\leq \sigma^2 T_{j,j}^{-2} \lambda_{\max}(\left(\boldsymbol{Q}^{\mathsf{H}}\boldsymbol{Q}\right)^{-1}), \tag{5}$$

for all $j \in [\![1, 3m + 2L]\!]$, or equivalently

$$\sigma^2 \lambda_{\max}^{-1}(\boldsymbol{Q}^{\mathsf{H}}\boldsymbol{Q}) \leq T_{j,j}^2 \boldsymbol{\Theta}_{\mathrm{CRB}}|_{j,j} \leq \sigma^2 \lambda_{\min}^{-1}(\boldsymbol{Q}^{\mathsf{H}}\boldsymbol{Q}) \tag{6}$$

for all $j \in [\![1, 3m + 2L]\!]$. To complete the proof, it remains to show that

$$\lambda_{\min}(\boldsymbol{Q}^{\mathsf{H}}\boldsymbol{Q}) \geq K_{\max}^{-1} \tag{7a}$$

$$\lambda_{\max}(\boldsymbol{Q}^{\mathsf{H}}\boldsymbol{Q}) \leq K_{\min}^{-1}, \tag{7b}$$

and one can obtained the desired bounds (1) by substituting (7) into (4) to conclude on the desired statement.

To show that (7) hold, we harness the confluent Vandermonde structure of the matrix $\boldsymbol{Q}$ [1] and provide an upper and lower bound its spectrum as a function of the separation criterion $N\delta$ through a Beurling–Selberg type approximation of the rectangle function by bandlimited functions presented in [2]. For convenience, denote $\boldsymbol{B} = \sqrt{N}^{-1}[\boldsymbol{A_f}, \boldsymbol{D}]$, $\boldsymbol{B'} = \sqrt{C_N}^{-1}[\boldsymbol{\Lambda A_f}, \boldsymbol{\Lambda D}]$, and define

$$\widetilde{\boldsymbol{Q}} = \begin{bmatrix} \boldsymbol{B'} & \boldsymbol{B} & \boldsymbol{0} \\ \overline{\boldsymbol{B'}} & \boldsymbol{0} & \overline{\boldsymbol{B}} \end{bmatrix} \in \mathbb{C}^{2N \times 3(m+L)}. \tag{8}$$

Since all the columns of $\boldsymbol{Q}$ are included in the columns of $\widetilde{\boldsymbol{Q}}$ one has

$$\lambda_{\min}(\boldsymbol{Q}^{\mathsf{H}}\boldsymbol{Q}) \geq \lambda_{\min}(\widetilde{\boldsymbol{Q}}^{\mathsf{H}}\widetilde{\boldsymbol{Q}}) = \sigma_{\min}^2(\tilde{\boldsymbol{Q}}) \tag{9a}$$

$$\lambda_{\max}(\boldsymbol{Q}^{\mathsf{H}}\boldsymbol{Q}) \leq \lambda_{\max}(\widetilde{\boldsymbol{Q}}^{\mathsf{H}}\widetilde{\boldsymbol{Q}}) = \sigma_{\max}^2(\tilde{\boldsymbol{Q}}). \tag{9b}$$

Equations (7) and (9) suggests that it is sufficient to show $\sigma_{\max}(\boldsymbol{Q}) \leq K_{\min}^{-1}$ and $\sigma_{\min}^2(\boldsymbol{Q}^{\mathsf{H}}\boldsymbol{Q}) \geq K_{\max}^{-1}$ to conclude which can be done by invoking [3, Theorem 5]. □

References

[1] W Gautschi, "On the inverses of vandermonde and confluent vandermonde matrices. i, ii," *Numer. Math*, vol. 4, pp. 117–123, 1962.

[2] Jeffrey D Vaaler, "Some extremal functions in fourier analysis," *Bulletin of the American Mathematical Society*, vol. 12, no. 2, pp. 183–216, 1985.

[3] Maxime Ferreira Da Costa, "The condition number of weighted non-harmonic Fourier matrices with applications to super-resolution," *hal-04261330*, 2024.